FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Revision Test 2026

Chemistry — Class 9 (Science Group)

Syllabus: Ch 1 — Nature of Chemistry | Ch 2 — Matter | Ch 3 — Atomic Structure | Ch 4 — Periodic Table & Periodicity | Ch 5 — Chemical Bonding | Ch 6 — Stoichiometry | Ch 7 — Electrochemistry | Ch 8 — Energetics | Ch 9 — Chemical Equilibrium

Total Marks: 31 **Time: 1 Hour 30 Minutes** **Paper: Exam-18 (Version 1)**

1. Attempt **ALL** questions. There are **no choices** in this paper.
2. Section A (12 marks): 12 MCQs. Fill the correct bubble. Suggested time: 15 minutes.
3. Show complete working for every numerical: formula, substitution, answer with correct units.
4. Use black or blue ink only. Pencil is allowed only for diagrams. Calculators are permitted.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Q.1 Choose the best possible option.	A	B	C	D	Answer
1	The branch of chemistry that designs products and processes so as to reduce or eliminate hazardous substances is:	Industrial chemistry	Green chemistry	Nuclear chemistry	Astrochemistry	ⒶⒷⒸⒹ
2	Liquid crystals are an intermediate state of matter that is commonly used in:	welding arcs	display screens (LCDs)	nuclear reactors	fire extinguishers	ⒶⒷⒸⒹ
3	The property of an element to exist in different physical forms in the same physical state is called:	isotopy	allotropy	sublimation	polymerisation	ⒶⒷⒸⒹ
4	The number of neutrons in an atom of ^{56}Fe (atomic number = 26) is:	26	30	56	82	ⒶⒷⒸⒹ
5	The discovery of atomic number in 1913, which became the basis of the modern periodic table, was made by:	Mendeleev	Moseley	Dalton	Rutherford	ⒶⒷⒸⒹ
6	The number of elements present in the **sixth** period of the periodic table is:	2	8	18	32	ⒶⒷⒸⒹ
7	During the formation of potassium oxide (K_2O), each potassium atom:	gains 1 electron	loses 1 electron	gains 2 electrons	loses 2 electrons	ⒶⒷⒸⒹ
8	The empirical formula of ethene, C_2H_4, is:	CH	CH_2	C_2H_4	CH_4	ⒶⒷⒸⒹ
9	For which one of these substances is the term **formula mass** used instead of **molecular mass**?	CO_2	NaCl	H_2O	CH_4	ⒶⒷⒸⒹ
10	When the equation ___ C_2H_2 + ___ O_2 → ___ CO_2 + ___ H_2O is balanced using the smallest whole numbers, the coefficient of O_2 is:	2	3	5	7	ⒶⒷⒸⒹ
11	The oxidation number of chlorine in $KClO_2$ is:	+1	+3	+5	+7	ⒶⒷⒸⒹ
12	For an **exothermic** reaction, which relationship between the enthalpy of the products and the reactants is correct?	$H_{products} > H_{reactants}$	$H_{products} < H_{reactants}$	$H_{products} = H_{reactants}$	ΔH is positive	ⒶⒷⒸⒹ

SECTION B — Short Questions (3 × 3 = 9 Marks)

Q.2	Attempt all three parts. Show your working.	Marks
(i)	(a) Calculate the molar mass of sulphur dioxide, SO_2. **[1]** (b) Hence calculate the mass, in grams, of **one molecule** of SO_2. **[2]**	3

	(S = 32, O = 16, $N_A = 6.02 \times 10^{23}$)	
(ii)	**Draw** the electron dot-and-cross structure of a water molecule (H_2O), showing only the valence electrons. Name the type of bond present between the atoms, and state how many **lone pairs** of electrons the oxygen atom has. (H = 1, O = 8)	**3**
(iii)	Determine the oxidation number of the **underlined** element in each of the following, showing your working: (a) <u>S</u> in H_2SO_3 (b) <u>C</u> in Na_2CO_3 (c) <u>I</u> in KIO_4	**3**

SECTION C — Long Questions (2 × 5 = 10 Marks)

Q.3	**Attempt both questions. Show complete working.**	**Marks**
(i)	A compound was analysed and found to contain **26.7% carbon, 2.2% hydrogen and 71.1% oxygen** by mass. **(a)** Determine the **empirical formula** of the compound. **[3]** **(b)** The molar mass of the compound is 90 g/mol. Determine its **molecular formula**. **[1]** **(c)** Calculate the number of **moles** present in 18 g of this compound. **[1]** (C = 12, H = 1, O = 16)	**5**
(ii)	**(a)** Silver occurs naturally as two isotopes: ^{107}Ag with an abundance of 52% and ^{109}Ag with an abundance of 48%. Calculate the **relative atomic mass** of silver, correct to two decimal places. **[2]** **(b)** State the number of **protons, neutrons and electrons** present in one atom of ^{109}Ag. (Atomic number of Ag = 47) **[1]** **(c)** Write the complete electronic configuration of nitrogen (atomic number = 7) in **s, p** notation, and state the **group** and **period** to which it belongs, giving a reason for each. **[2]**	**5**

Invigilator's Signature: ______________________ Candidate's Signature: ______________________

Chapters Covered: Chapters 1 to 9

— End of Question Paper —